Installing Bump'em on a lab PC

What this does

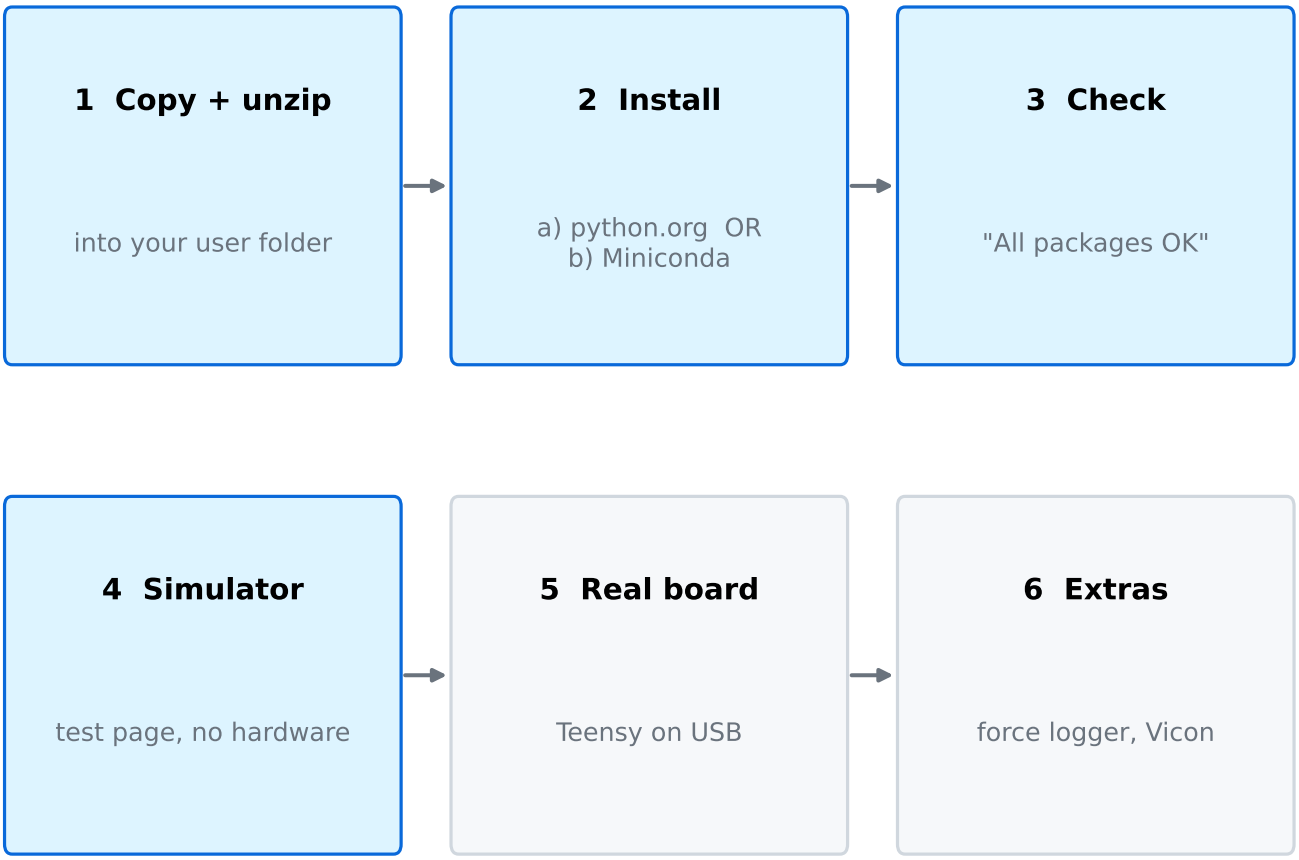
Puts the Bump'em server, the test page and the force logger on a Windows 10/11 PC that has NO internet (e.g. the Vicon PC). Everything installs for your Windows user only: no admin rights, nothing in C:\Program Files, no system settings changed. It can be removed completely in 2 minutes (last page).

What you need

- the bundle folder BumpemLab_<date> (or its .zip) on a USB stick
- optional, for the Vicon steps: the Vicon DataStream SDK installer (vicon.com, needs an account)
- about 20 minutes, the Teensy USB cable for step 5

Before you start

- **Ask the PC's administrator first: is there app lockdown or a write filter? (installs may be blocked or vanish at reboot). Do not change ESET or any security setting yourself.**
- Do it when nobody is recording on that PC.

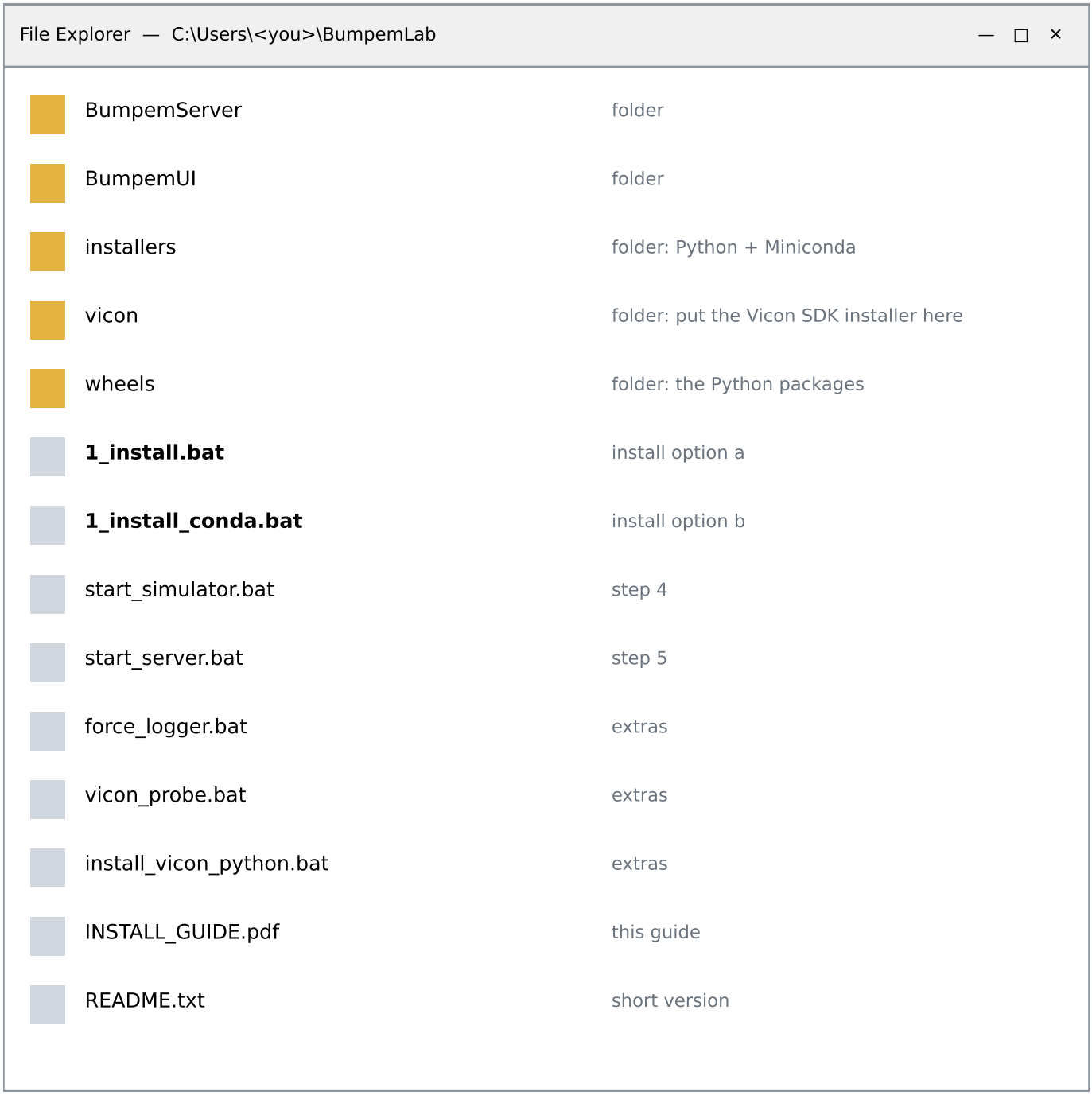


Step 1: copy the folder to the PC

1. Plug in the USB stick. Open File Explorer.
2. If you have the .zip: right-click it > Extract All... > choose C:\Users\<you> > Extract.
   If you have the folder: copy it to C:\Users\<you> (Ctrl+C, Ctrl+V).
3. Rename the folder to BumpemLab (optional, shorter).
4. If you have the Vicon SDK installer: copy it into the vicon folder inside.

Install it where it will stay. After step 2 the folder must not be moved or renamed.

Do not run anything from inside the .zip or from the USB stick.



Step 2: install, choose ONE of the two

Both install Python 3.12 and the same packages, offline, for your user only. Pick one and only one.

a) 1_install.bat (recommended if you don't need conda)

Python from python.org + a small environment inside the BumpemLab folder. Smaller, fewer steps.

b) 1_install_conda.bat (if you want conda, like on the lab Mac)

Anaconda's Miniconda + a conda environment called BumpemHome. You must accept Anaconda's Terr
of Service (it asks). Free for organisations under 200 employees: IBMS has 6.

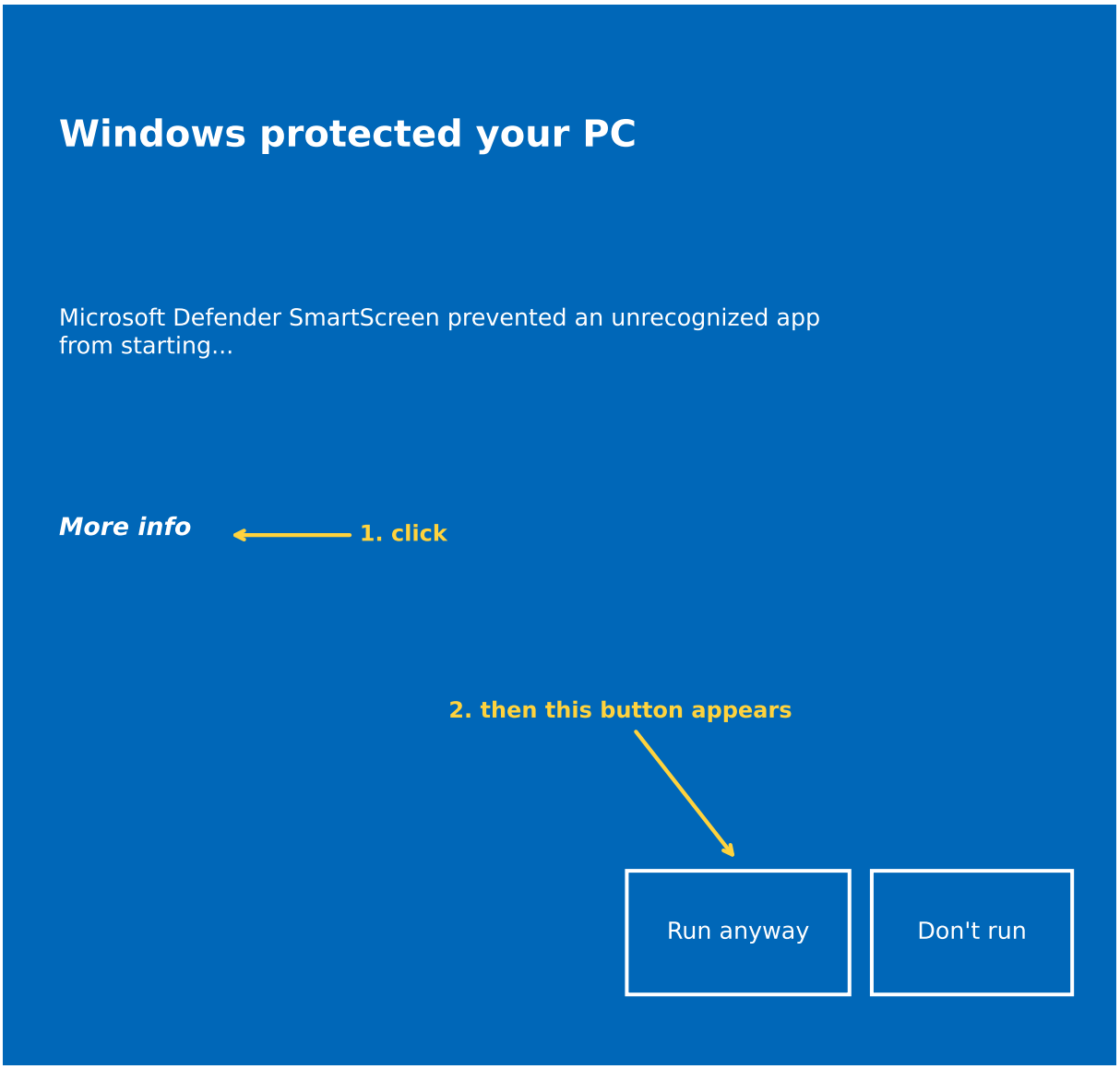
How to run a .bat file

Double-click it in File Explorer. A black window opens and shows what happens. Read the last lines.

If Windows shows a blue box "Windows protected your PC"

Click More info > Run anyway (these are your own files from the bundle).

If there is no "Run anyway", or ESET blocks it: stop and ask the administrator. Do not change settings.



Step 2a: 1_install.bat

1. Double-click 1_install.bat. Wait: Python installs silently (about 1 minute, nothing to click).
2. Then the packages install from the wheels folder (about 30 s).

Done when the last lines are:

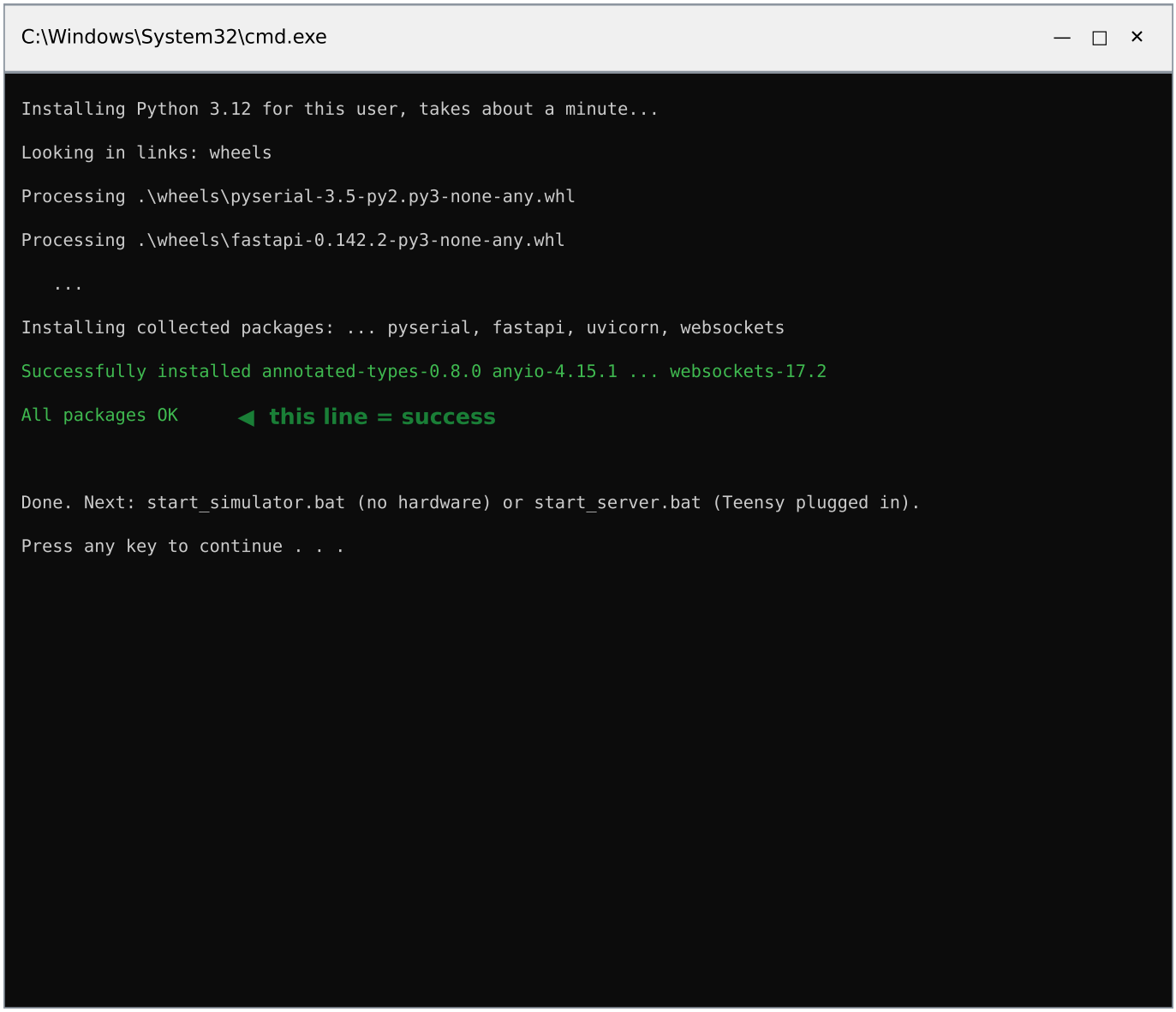
All packages OK

Done. Next: start_simulator.bat ...

3. Press any key to close the window. Go to step 4.

If it stops with a message instead

- "Python did not install": run installers\python-3.12.10-amd64.exe by hand: tick "Install for this user only" if offered, untick "Add python.exe to PATH", Install. Then run 1_install.bat again.
- "package install failed": photograph the window (all red lines) and send it to Claude.
- Running it twice is safe: it skips what is already installed.



Step 2b: 1_install_conda.bat

1. Double-click 1_install_conda.bat.
2. It shows the link to Anaconda's Terms of Service and asks: Do you accept ... [Y,N]?

   Press Y to accept and continue, N to stop (then use 1_install.bat instead).
3. Miniconda installs silently (a few minutes, nothing to click), then the conda environment BumpemHome is created and the packages are installed into it.

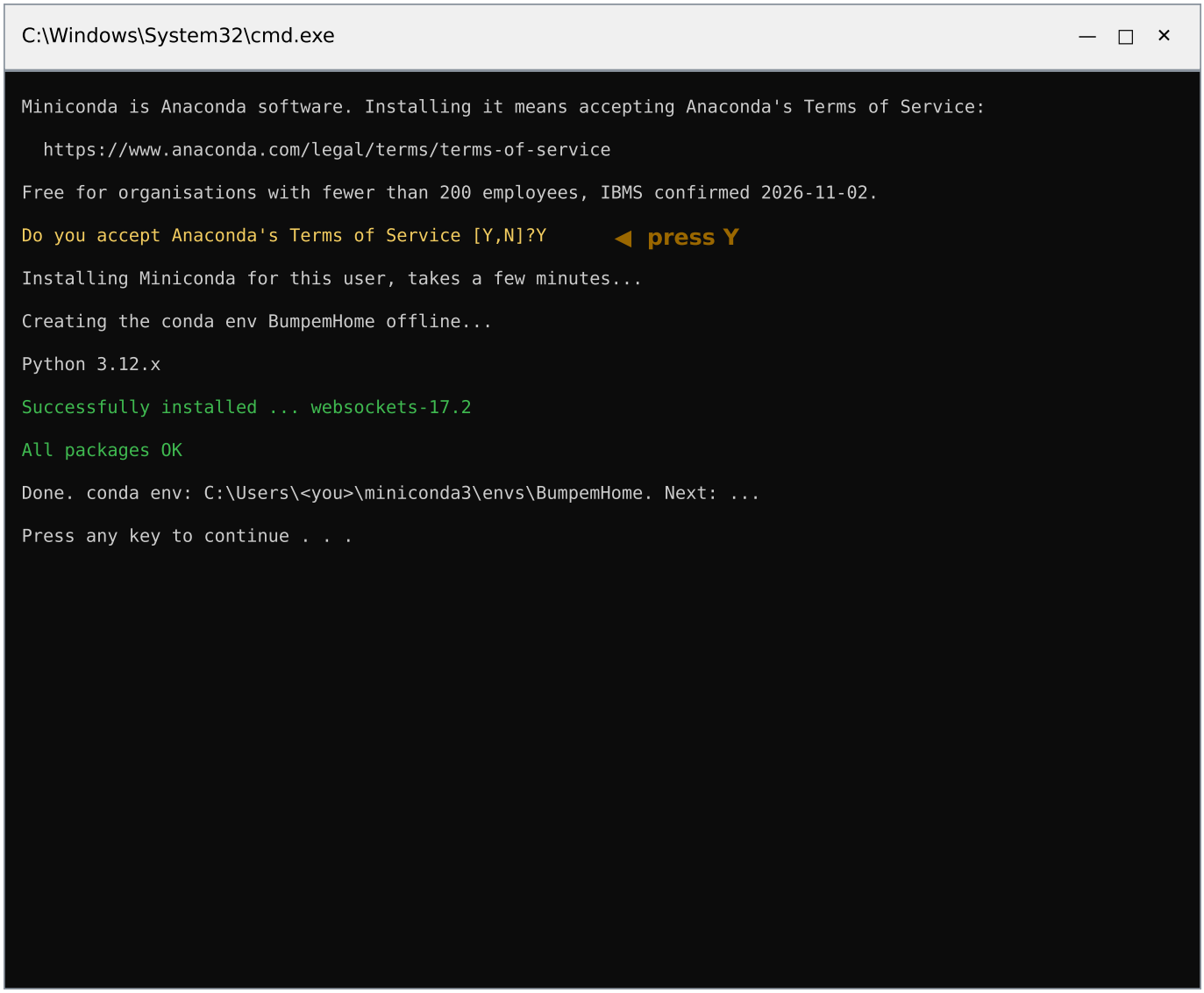
Done when the last lines are: All packages OK and Done. conda env: ...

Use the environment by hand (optional)

Start menu > Anaconda Prompt > conda activate BumpemHome (the .bat files find it by themselves).

If it stops with a message

- "Miniconda did not install": run the Miniconda installer in installers by hand: Just Me, untick "Add to PATH" and "Register as default Python". Then run 1_install_conda.bat again.
- "Could not create the conda env" or "not Python 3.12": photograph the window, send it to Claude, and use 1_install.bat meanwhile.



Step 4: test with the simulator (no hardware)

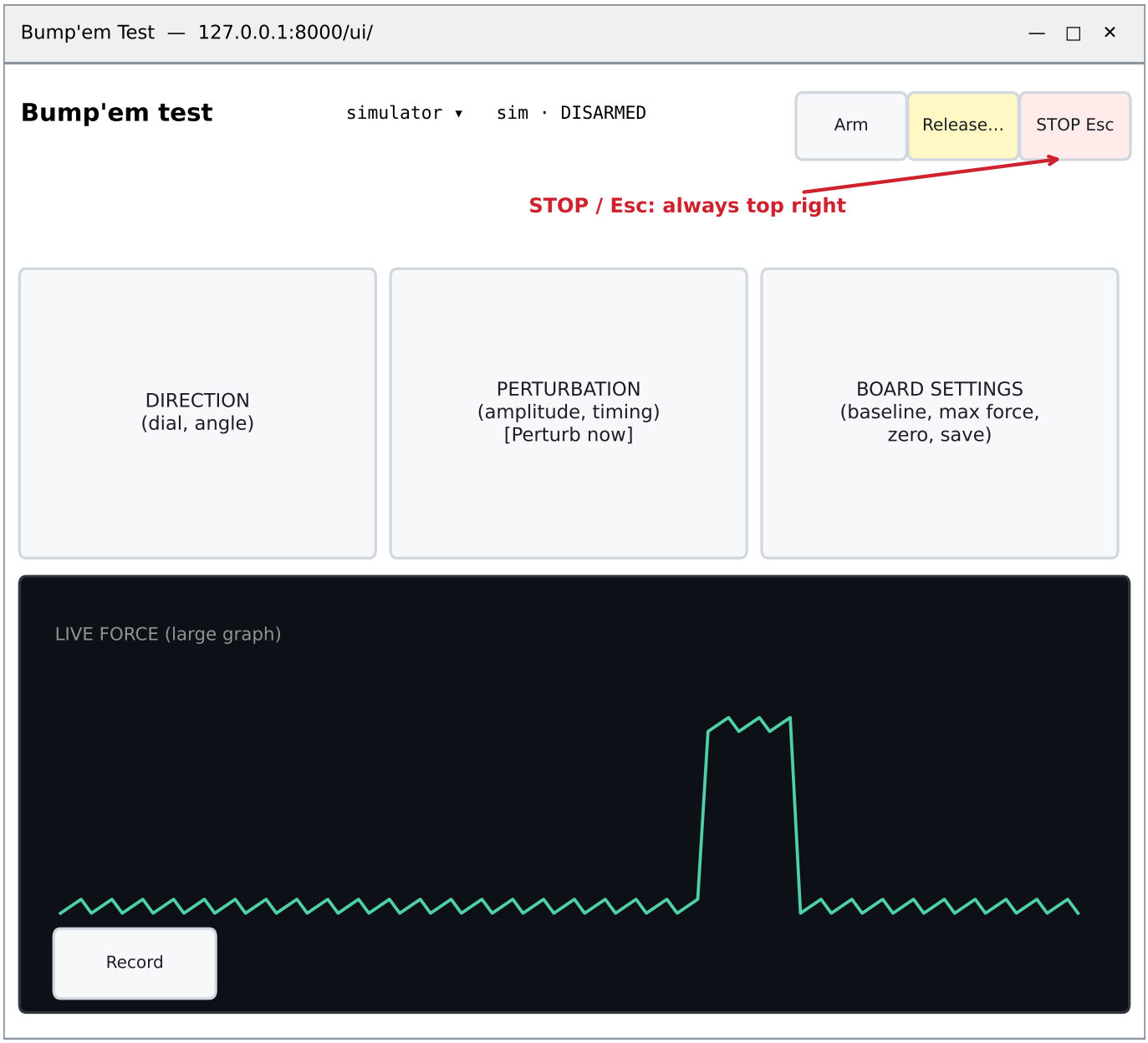
1. Double-click start_simulator.bat. A black window opens and stays open: that IS the server.
2. After about 4 s the browser opens http://127.0.0.1:8000/ui/ by itself (if not, type that address).
3. Top right shows sim and DISARMED. Try it:

   Record > Arm > set Angle and Amplitude > Perturb now > watch the graph > Stop recording.

   The force CSV downloads to your Downloads folder.
4. To stop: close the black window.

Nothing here talks to the Teensy or the motors. Saved settings and logs go to the sim folder.

If the black window closes at once or shows red text: photograph it and send it to Claude.

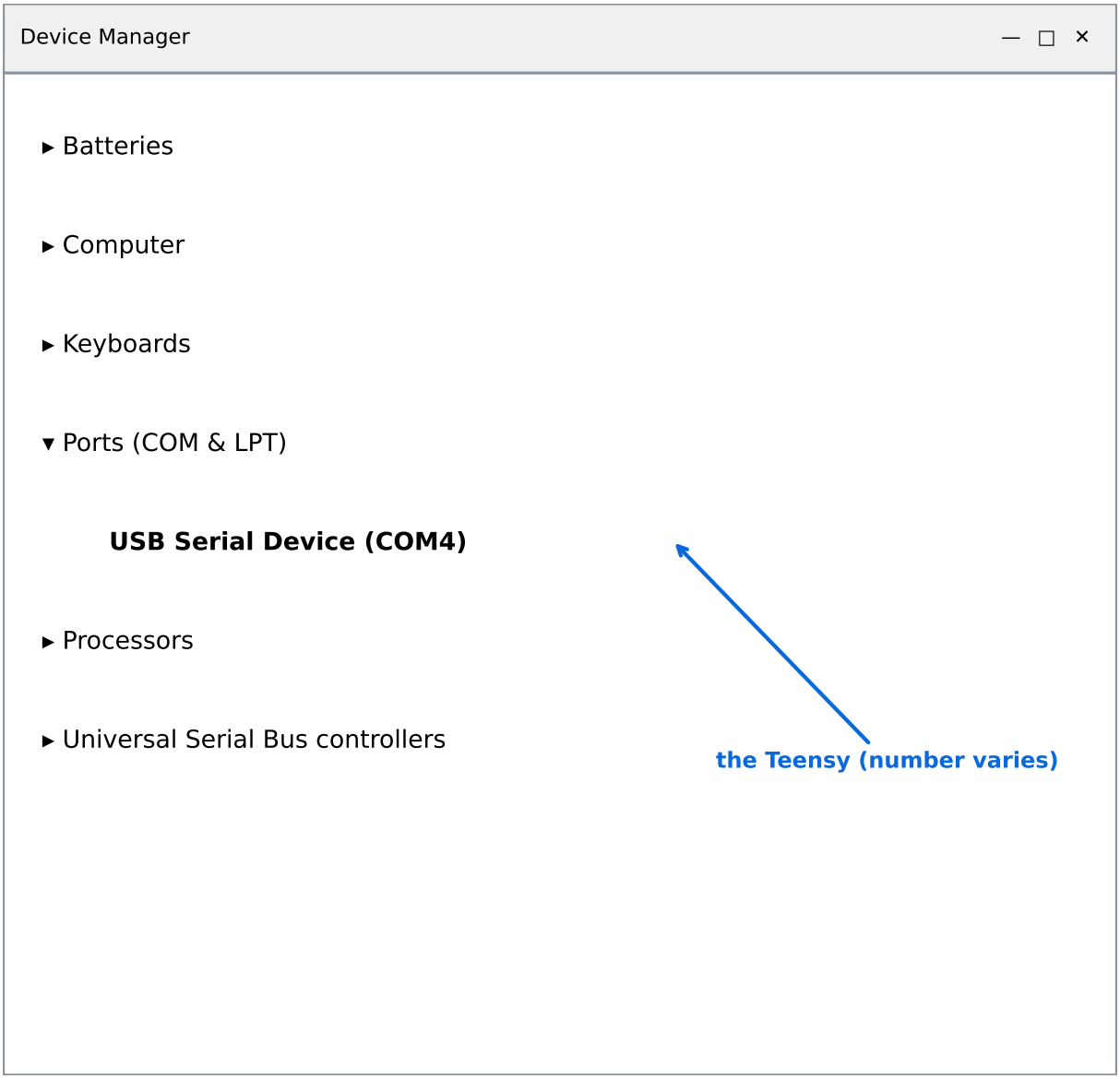


Step 5: with the real board

1. Close the simulator window if it is open (only one server at a time).
2. Plug the Teensy into the PC's USB. Windows needs no driver for it.
3. Find its COM port: right-click Start > Device Manager > Ports (COM & LPT): a USB Serial Device (COMx). Unplug and replug the Teensy: the line that disappears and comes back is the Teensy.
4. Double-click start_server.bat. The page opens. Choose that COM port in the list, press Connect. The lab's saved settings (calibration, gains, max force) are written to the board at every connect.
5. Same page as on the Mac. Recordings: Downloads + BumpemServer\Logs.

Safety (same rules as on the Mac)

- **STOP button or Esc = drivers off at once, ropes go slack. Stop the treadmill first unless it is an emergency.**
- **Arm only with the 48 V supplies on and nobody in the ropes until you are ready.**
- If the USB link drops: replug the Teensy, then Disconnect + Connect in the page.



Step 6: extras

Force logger (one load cell, live plot, CSV)

1. Close the server window (the logger needs the Teensy for itself).
2. Double-click force_logger.bat (tower D). Other tower: open a command prompt in the folder and type:

   force_logger.bat --tower B

3. First time, and after moving the sensor to another amplifier: press Calibrate and follow the two steps with the 12 kg weight. With the weight on it must read about 118 N / 12.00 kg.
4. Start > Stop > Save CSV. Threshold line: type a force in the box under the graph.

Vicon (Nexus must be open in Live mode)

1. Install the Vicon DataStream SDK from the vicon folder (its own installer; may need the administrator).
2. Find the SDK's Python folder (inside the SDK install, a folder named Python containing vicon_dssdk).
3. Open a command prompt in BumpemLab (in File Explorer: click the address bar, type cmd, Enter), then:

   install_vicon_python.bat "C:\Program Files\Vicon\DataStream SDK\...\Python"

   Done when it prints vicon_dssdk OK.

4. vicon_probe.bat (on the Nexus PC itself). From another PC: vicon_probe.bat 192.168.1.60:801
5. Stand on one belt during the 10 s: note which plate is the LEFT belt. Send the output to Claude.

If something goes wrong, and how to remove it

What you see	What to do
Blue box "Windows protected your PC"	More info > Run anyway. No "Run anyway" button: ask the administrator.
ESET warning / file quarantined	Do not change ESET. Tell the administrator which file; ask them to allow it.
Installed, but gone after a reboot	The PC has a write filter. The administrator must allow or commit the install.
"Not installed yet: run 1_install..."	Step 2 was not done, or the folder was moved after it. Run step 2 again.
Page does not open	Wait 5 s, then type 127.0.0.1:8000/ui/ in the browser. The black window must be open.
Black window: "address already in use"	Another server is still running: close all black windows, try again.
"COMx is busy" / Access is denied	Another program has the Teensy: close the logger, other servers, ESCON tools.
No COM port in the list	Other USB cable or port (some cables are charge-only). Check Device Manager.
"no Teensy found" (force logger)	Teensy not plugged in, or the server holds it: close the server.
Values look wrong (force logger)	Calibrate again with the 12 kg weight; 24 V of that tower on?

Asking for help

Photograph the whole black window (or copy its text: right-click the title bar > Edit > Select All, Enter) and send it to Claude together with which step you were on.

Remove everything (2 minutes, puts the PC back as it was)

1. Settings > Apps > "Python 3.12.10" > Uninstall (if you used 1_install.bat)

   Settings > Apps > "Miniconda3" > Uninstall (if you used 1_install_conda.bat; also removes BumpemHome)

2. Delete the BumpemLab folder.